



AKTU

B.Tech II-Year



Cyber security



Unit-3 in One Shot

Pdf Notes + PYQs + Imp. Questions

AKTU IMP QUESTIONS

1.	Explain 7 tools used in cyber crime	AKTU 2023-24
2.	What is identity theft ? How it is done ? How identity theft can be handled	AKTU 2023-24
3.	What is steganography ? Explain in detail	AKTU 2023-24
4.	Explain DOS and DDOS attack and their potential consequences	AKTU 2024-25
5.	Explain the use of proxy server and anonymizer in cyber crime	AKTU 2024-25
6.	Describe the methods used to exploit buffer overflow	AKTU 2024-25
7.	difference between DOS and DDOS attack?	AKTU 2023-24
8.	What is SQL injection? How does it exploit vulnerabilities in website	AKTU 2023-24
9.	How botnet activities in disrupted by cyber professionals	AKTU 2023-24
10.	What is steganography? How it is used to conceal information	AKTU 2023-24
11.	What is a keylogger? How does it capture the keystroke?	AKTU 2023-24

Proxy Server

- A proxy server is an **intermediate server** that places between a **user's device** and the internet.
- It receives requests from the user (like to access a website), forwards them to the target server, and then sends the response back to the user.
- It hides the user's real IP address and can control or filter internet traffic.

ADVANTGES:

Advantage	Explanation
1. Privacy Protection	Hides the user's IP address and identity while browsing.
2. Access Control	Organizations can control employee access to certain websites.
3. Content Filtering	Blocks unwanted content like ads or harmful websites.
4. Caching	Speeds up access to frequently visited websites by storing cached versions.

Disadvantages:

Disadvantage	Explanation
1. Security Risks	If poorly configured, it can leak data or be used for malicious activities.
2. No Encryption	Most proxies don't encrypt data
3. Limited Anonymity	Not all proxies fully hide the user's activity or location.
4. Slower Speed	Can cause delays if overloaded or improperly configured.
5. Incompatibility	Some websites detect and block proxy server traffic.

Anonymizer

- An anonymizer is a tool (usually a website, software, or server) that helps you hide your **identity** while using the internet.
- It hides your IP address and makes your online activity anonymous.
- It works like a shield between you and the websites you visit—so the websites don't know who you are or where you're from.

Reasons for Using Anonymizer:

Reason	Explanation
1. Privacy Protection	Keeps your personal data and location hidden.
2. Avoid Tracking	Stops websites or ads from tracking your online behavior.
3. Stay Safe on Public Wi-Fi	Prevents hackers from spying on you on open networks.
4. Secure Communication	Keeps your messages or browsing safe from spying.

Point	Proxy Server	Anonymizer
Main Purpose	Acts as a middleman to control and manage internet use	Hides user identity to ensure complete online anonymity
Privacy Level	Provides some privacy (IP hiding)	Provides high privacy and anonymity
User Control	Often used by organizations (e.g., schools, offices)	Used by individuals for private and anonymous browsing
Content Filtering	Can block or allow specific websites	Usually does not filter content

Use Proxy Server in cybercrime

Use	Explanation
Hide Real IP Address	Attackers use proxy servers to hide their location during cyberattacks.
Bypass Security Systems	Proxies help avoid IP-based blocks, firewalls, or restrictions.
Control Botnets	Cybercriminals route commands to infected computers using proxy servers.
Phishing Attacks	Proxies are used to host fake login pages while keeping the attacker's identity hidden.

Use Anonymizer in cybercrime

Use	Explanation
High-Level Anonymity	Criminals use anonymizers to hide identity from law enforcement.
Access Dark Web	Anonymizers are used to access illegal websites on the dark web.
Data Theft & Communication	Stolen data is shared anonymously using anonymizers to avoid tracking.
Illegal Transactions	Criminals use anonymizers to trade drugs, weapons, or stolen data anonymously.

What is Phishing?

Phishing is a type of cyber attack where attackers pretend to be someone trustworthy (like a bank, company, or friend) to trick you into giving away personal information, like:

Passwords, Credit card numbers

Bank details, OTPs (One-Time

Passwords

How Phishing Works

Bait Sent: The attacker sends a fake email, message, or link (looks real).

User Clicks: Victim clicks the link or opens the message.

Fake Page Opens: It looks like a real login page (bank, email, etc.).

User Enters Details: The victim types in their password or info.

Info Stolen: The attacker gets that data and misuses it.

Types of phishing attack

Type of Phishing Attack	What It Means	Example
Email Phishing	Fake emails pretending to be from a real company or person.	You get an email saying your bank account will be closed unless you click a link.
Spear Phishing	Targeted phishing aimed at a specific person or group.	A fake email sent only to your office team, pretending to be your boss asking for passwords.
Smishing (SMS Phishing)	Phishing done through text messages on your phone.	You receive a text asking to verify your mobile number by clicking a suspicious link.

Types of phishing attack

Type of Phishing Attack	What It Means	Example
Vishing (Voice Phishing)	Phishing using phone calls or voice messages to trick you.	Someone calls pretending to be your bank and asks for your ATM PIN.
Clone Phishing	A real email is copied, but the link or attachment is replaced with a malicious one.	You get a copy of an earlier email from a friend but the attachment has malware.
Pharming	Redirecting you from a real website to a fake one without you knowing.	You type your bank website URL but are redirected to a fake site that steals your info.
Whaling	Phishing targeting big important people like CEOs or executives.	A fake email sent to the company CEO pretending to be from a trusted source asking for confidential info.

Password Cracking

- Password Cracking is the process of recovering or guessing someone's password without permission using different methods or tools.
- Hackers use password cracking to break into user accounts, systems, or networks by finding out the correct password.

Purpose	Explanation
Unauthorized Access	To get into a user's email, bank, or social media without permission.
Stealing Data	To take private or sensitive information like photos, messages, or money.
Testing Security (Ethical)	Security experts use it to check system strength (ethical hacking).
Spying or Blackmailing	Criminals use cracked passwords to spy, control, or blackmail users.
Spreading Malware	Attackers may install viruses after getting access.

Classification of password attack

Online Password Cracking Attacks

➤ In **online attacks**, the attacker tries to guess the password **directly on a live system**, like a website or email login page.

Technique	Explanation
1. Brute Force Attack	Tries all possible passwords directly on the login page until one works.
2. Dictionary Attack	Tries common or popular passwords (like 123456, admin, qwerty) online.
3. Phishing	Sends fake login pages to trick users into typing their real passwords.
4. Social Engineering	Tricks or manipulates the user (e.g., “I’m your bank. Tell me your password.”)
5. Credential Stuffing	Uses leaked username-password combinations from other sites to try logging in.

Classification of password attack

offline Password Cracking Attacks

➤ In **offline attacks**, the attacker steals or copies password data (like password hash files) and tries to crack them **without connecting to the system**.

Technique	Explanation
1. Brute Force Attack	Tries all possible combinations on stolen password files (e.g., hashes).
2. Dictionary Attack	Uses a file of common passwords to match against stolen password hashes.
3. Rainbow Table Attack	Uses pre-computed tables to quickly find a matching hash and reveal the password.
4. Hybrid Attack	Mixes dictionary words with numbers/symbols (e.g., pass123, admin2024).

What is a Keylogger?

- A keylogger (short for keystroke logger) is a spyware tool or program that records every key you press on your keyboard — without you knowing.
- It is used by:
 - Hackers to steal passwords, messages, or personal data
 - Security researchers or employers to monitor computer activity (in legal cases)

How Does a Key logger Capture Keystrokes?/

Types of key loggers

Keyloggers can be software-based or hardware-based.

Software Keylogger (Most Common)

- Installed secretly on your system (often via malware, fake apps, or phishing). Runs in the background.
- Records everything you type (passwords, messages, credit card numbers).
- Saves it in a file and sends it to the attacker.

Example:

You type: mybankpassword123

The keylogger saves and sends: m, y, b, a, n, k... etc.

Hardware Key-logger

- A physical device connected between your keyboard and computer (usually USB).
- Captures keystrokes directly from the keyboard.
- Doesn't need software and works even without internet.

What is an Anti-Keylogger?

- An anti-keylogger is a security tool or software designed to detect, block, or remove keyloggers from your device. It protects you from spyware that tries to record your keystrokes (like passwords, PINs, or messages).
- It works like an antivirus, but it's specially focused on stopping keylogging attacks

Advantage of anti-keylogger	Explanation
Detects Hidden Key loggers	Finds and alerts you about hidden software trying to spy on your typing.
Protects Sensitive Info	Stops your passwords, credit card numbers, and chats from being stolen.
Removes Malware	Can remove harmful key logger programs from your device.
Secures Online Activity	Helps keep your online banking, email, and logins safe from spying.
Can Be Used With Antivirus	Often works alongside antivirus software for full protection.

SPYWARE

Spyware is a type of **malicious software (malware)** that secretly **monitors and collects information** from your computer or mobile device **without your knowledge**.

Type	Explanation
1. Keylogger	Records everything you type, like passwords, chats, and credit card numbers.
2. Adware	Shows unwanted ads and pop-ups, and tracks your browsing to display targeted ads.
3. Trojan	Looks like a safe program but secretly installs spyware or gives control to hackers.
4. Tracking Cookies	Small files that track your activity across websites.
5. Password Stealer	Designed to find and send your saved passwords
6. Infostealer	Collects various personal data like documents, login info, or screenshots.

What is a Computer Virus?

A **computer virus** is a harmful program that **spreads from one file or computer to another**. It can **damage files, slow down your system, or steal information**.

Type	What it Does
File Virus	Attaches to program files and spreads when opened.
Boot Virus	Infects the startup part of the system (boot sector).
Macro Virus	Hides in Word/Excel files and spreads when opened.
Resident Virus	Stays in memory and infects files in the background.
Multipartite Virus	Attacks both boot and file areas to spread faster.
Polymorphic Virus	Changes its form to avoid antivirus detection.
Email Virus	Spreads through email attachments or links.

What is a Computer Worm?

- A computer worm is a type of malicious software that spreads itself automatically from one computer to another without needing to attach to any file.
- Unlike a virus, a worm doesn't need human help to spread — it moves on its own through networks (like Wi-Fi or the internet).

Key Features of a Worm :

- Spreads by itself (no need to click or open a file)
- Uses networks to travel (emails, file sharing, etc.)
- Can slow down or crash systems by using too much memory or bandwidth
- May be used to install other malware, steal data, or open backdoors

Difference Between Computer Virus and Worm

Point	Computer Virus	Computer Worm
Spreads How?	Needs user action (like opening a file) to spread	Spreads automatically through networks
Needs a Host File?	Yes, attaches to a file or program	No, it is standalone
User Interaction?	Required (e.g., clicking a file)	Not required
Speed of Spread	Slower	Faster (self-replicating)
Main Purpose	Often damages files or programs	Spreads quickly, may also open the system for other attacks
Example	Infected Word file, game, or software	Worm spreading vai local network

7 tools used in cyber crime

Tool Name	What It Does
1. Keylogger	Records everything a person types (like passwords or messages) secretly.
2. Phishing Kits	Ready-made fake websites/emails to trick users into giving personal info.
3. RAT (Remote Access Trojan)	Gives full control of your computer to the hacker without your permission.
4. Spyware	Secretly watches your activity and steals data like files, passwords, or location.
5. Password Cracking Tools	Guess or crack passwords using techniques like brute force (e.g., John the Ripper).
6. Botnets	A network of infected computers used to send spam or launch attacks
7. Malware Builders	Help hackers create viruses, worms, or Trojans easily (no coding skills needed).

What is a Trojan Horse?

- A Trojan Horse (or Trojan) is a type of malware that pretends to be a safe or useful program, but once installed, it secretly performs harmful actions on your device.

Threat	Explanation
1. Data Theft	Steals your personal files, passwords, or banking information.
2. Remote Access	Allows hackers to control your device from anywhere
3. Spyware Activity	Watches your activity, records keystrokes, and takes screenshots.
4. Downloading More Malware	Installs other malware like ransomware, worms, or viruses.
5. System Damage	Deletes, modifies, or corrupts important files and programs.

What is a Backdoor?

- A backdoor is a hidden way to access a computer system, network, or software without the user knowing.
- It allows hackers (or sometimes even developers) to bypass normal security controls like passwords and firewalls.

Function	Explanation
Unauthorized Access	Allows hackers to enter a system without login or permission.
Remote Control	Lets attackers control the device from another location.
Data Theft	Used to steal files, passwords, or other sensitive information.
Install More Malware	Can be used to install ransomware, spyware, or viruses.
Disable Security Tools	Turns off antivirus or firewall to make the system more vulnerable.

How to Protect Your System from Trojan Horse and Backdoor Attacks

1. Install Antivirus/Anti-Malware Software

- Use trusted antivirus (e.g., Kaspersky, Norton, Bitdefender).
- Keep it updated regularly.
- Enable real-time protection to detect Trojans or backdoors.

2.Keep Your System and Software Updated

- Install OS updates and security patches regularly.
- Updates fix security holes that Trojans/backdoors can exploit.

3. Avoid Downloading from Unknown Sources

- Don't install cracked software, fake games, or apps from unknown websites.
- Use official stores (e.g., Microsoft Store, Google Play).

4.Be Careful with Email Attachments and Links

- Don't open links or files from unknown or suspicious emails.
- Phishing emails often spread Trojans or backdoor tools

5. Use a Firewall

- A firewall monitors incoming and outgoing traffic.
- It helps block unauthorized access and backdoor connections.

What is Steganography?

- Steganography is the method of hiding secret information inside something that looks normal, like a picture, audio, or video — so that no one knows the information is even there.

Uses of Steganography:

- Secret communication
- Watermarking
- Hiding files in cybercrime
- Secure sharing of confidential data

What is Steganalysis?

Steganalysis is the process of detecting, analyzing, and extracting hidden data from steganographic files.

Point	Steganography	Steganalysis
Meaning	Hiding secret information inside normal files (like images).	Detecting and extracting hidden information from those files.
Purpose	To send secret data without anyone noticing.	To find out if a file contains hidden data.
Used By	People who want to send information secretly.	Security experts, cyber analysts, or investigators.
Example	Hiding a message in an image.	Checking an image to see if it has a hidden message.
Goal	Secrecy.	Detection and analysis.

How Steganography Is used to conceal the information?

1. Image Steganography

- Secret data is hidden by changing some pixels in the image.
- Example: Slightly changing the last bit of color values (called LSB – Least Significant Bit).

2. Audio/Video Steganography

- Data is hidden by changing sound waves or frames slightly.
- It's not noticeable to the ear or eye but can store information inside.

3. Text Steganography

- Hides data using spaces, capital letters, font colors, or invisible characters.
- Example: Each capital letter in a paragraph could be part of a hidden message.

4. File Steganography

- Secret files are merged into other files (like hiding a ZIP inside a JPEG).
- The file works normally, but contains an extra hidden file inside.

Tools Used in Steganography

Tool Name	Media Type Supported	Key Features
OpenStego	Images	Hides data, supports watermarking
SilentEye	Images, Audio	User-friendly, supports encryption
Steghide	Images, Audio (e.g., JPEG, WAV)	Password protection, strong encryption
QuickStego	Images (BMP)	Simple tool for beginners, hides text
Xiao	Images (BMP), Audio (WAV)	Easy data hiding and extraction
DeepSound	Audio (WAV, FLAC)	Hides files in songs, supports stego audio CDs

Tools Used in Steganalysis

Tool Name	Function	Media Analyzed
StegExpose	Detects steganographic content using statistical methods	Images (JPEG, PNG)
StegSecret	Analyzes and extracts hidden data	Images
StegDetect	Detects steganography in JPEG images	Images (JPEG)
OpenStego (Analyze)	Detects watermarking and hidden data	Images
zsteg	Detects LSB steganography in PNG and BMP files	Images (PNG, BMP)
Binwalk	Analyzes binary files and extracts hidden content	Any binary file

DOS ATTACK

A DoS (Denial of Service) attack is a cyberattack where a hacker tries to make a website, server, or network unavailable by flooding it with too much traffic or requests, causing it to crash or slow down.

What is a DDoS Attack?

DDoS (Distributed Denial of Service) Attack is a type of cyberattack where multiple computers or devices (called a botnet) are used to send massive traffic to a website, server, or network to crash it or make it unavailable to real users.

Types/Levels of DoS Attacks

Attack Name	Type of Attack	How It Works	Effect on Target
1. Flood Attack	Volume-Based	Sends a huge amount of fake traffic or requests to overload the server or network.	Server becomes slow or crashes due to overload.
2. Ping Attack (ICMP Flood)	Volume-Based	Sends too many ping (ICMP echo) requests to the target.	Wastes bandwidth and makes the system unresponsive .
3. SYN Flood Attack	Protocol-Based	Sends fake TCP connection requests (SYN) but never completes the connection.	Server waits and wastes resources, causing a crash or freeze .
4. Teardrop Attack	Protocol-Based (IP Fragmentation)	Sends broken/overlapping data packets that the system can't reassemble properly.	Causes system to confuse and crash .

Point	DoS (Denial of Service)	DDoS (Distributed Denial of Service)
Full Form	Denial of Service	Distributed Denial of Service
Number of Attackers	Single system or attacker	Multiple systems (botnet of infected devices)
Attack Source	One computer or IP address	Many computers from different locations
Power of Attack	Less powerful	More powerful and harder to stop
Detection	Easier to detect and block	Harder to trace due to multiple sources
Speed of Attack	Slower	Faster and more damaging
Example	One PC flooding a website with traffic	Thousands of infected PCs flooding a website at the same time

Potential Causes of DoS and DDoS Attacks

Cause	Explanation	Applies To
1. Exploiting Vulnerabilities	Attackers find and exploit weaknesses in servers, applications, or network protocols.	DoS & DDoS
2. Botnets (For DDoS)	Large groups of infected devices (computers, IoT) controlled remotely to launch coordinated attacks.	DDoS only
3. Overloading Network Resources	Sending huge amounts of traffic to overwhelm bandwidth, CPU, or memory.	DoS & DDoS
4. Poorly Configured Networks/Devices	Misconfigured routers or firewalls can be exploited to amplify attacks (e.g., DNS amplification).	DoS & DDoS
5. IP Spoofing	Fake source IP addresses hide attackers and help send reflected traffic to the victim.	DoS & DDoS
6. Lack of Security Measures	Absence of firewalls, intrusion detection, or rate limiting makes systems easy targets.	DoS & DDoS
7. Motivations of Attackers	Financial gain (ransom), revenge, competition, or just causing disruption.	DoS & DDoS

What is SQL Injection?

SQL Injection (SQLi) is a type of cyber attack where an attacker inserts (injects) harmful SQL commands into a website's input fields (like login forms or search boxes) to access or modify the database.

How SQL Injection Exploits

Website Vulnerabilities

Step	Explanation
1. Input Field Exists	Website has an input field (e.g., login, search) that connects to a database.
2. No Input Validation	The website does not check or clean the user input properly.
3. Malicious SQL is Entered	Attacker types SQL code into the field
4. Database Executes It	The website sends the attack code directly to the database.
5. Unauthorized Access or Data Theft	The attacker can view, delete, or change sensitive data (like passwords).

What SQL Injection Can Do

- Bypass login systems
- Access confidential data (like user emails, passwords, credit card info)
- Delete or modify database content
- Dump entire databases
- Gain admin-level access

SQL Injection Prevention

Countermeasure	Explanation
1. Input Validation	Always check and filter user input to allow only expected characters or formats .
2. Use Prepared Statements / Parameterized Queries	Use secure SQL coding methods where user input is treated as data, not code .
3. Stored Procedures	Use predefined SQL queries that accept only safe inputs and reduce risk.
4. Escaping User Input	Special characters (like ', --) should be escaped so they can't change the SQL query .
5. Use ORM Tools	Frameworks like Hibernate, Django ORM help write database queries more safely.

Countermeasure	Explanation
6. Least Privilege Access	Give your database only the needed permissions —don't allow full access for every user.
7. Error Message Control	Don't show raw database errors to users; it can give clues to hackers.
8. Web Application Firewall (WAF)	WAFs can block common SQLi patterns before they reach your server.
9. Regular Code Review & Testing	Scan your website regularly with security tools to find and fix SQLi vulnerabilities.
10. Use HTTPS	Ensures that data isn't modified during transit , reducing chances of injection.

What is a Blind SQL Injection Attack?

Blind SQL Injection is a type of **SQL Injection attack** where the attacker cannot see the actual database error messages or output — but still gains information by **observing the application's behavior or response**.

Type	Explanation
Boolean-Based	Sends queries that return true/false and observes the page behavior.
Time-Based	Sends queries that delay response if true (e.g., SLEEP(5)) to extract data.

What is Buffer Overflow?

- A Buffer Overflow is a programming error where a program writes more data into a memory buffer than it can hold
- This extra data overwrites adjacent memory, which can crash the program or let attackers run malicious code.

Type of buffer flow attack	Explanation
1. Stack Overflow Attack	An attacker feeds more data into a program's stack space than it expects. This extra data overwrites the saved return address, so when the function ends, it "returns" into attacker's code instead of the real code.
2. Heap Overflow Attack	Overwrites heap memory (used for dynamic memory) to change program behavior.
3. Format String Attack	Uses %s, %x, etc., in input to read/write memory locations.
4. Integer Overflow	Forces integer values to wrap around , leading to buffer overflow indirectly.
5. NOP Sled Attack	Uses a sequence of NOP (No Operation) instructions to increase chances of executing malicious code.

Prevention techniques

Description

Bounds Checking

Always check the size of data before copying to buffers.

Use Safe Functions

Use strncpy() instead of strcpy(), snprintf() instead of sprintf().

Stack Canaries

Use special values in memory to detect overflow before executing code.

ASLR (Address Space Layout Randomization)

Makes memory addresses unpredictable.

DEP (Data Execution Prevention)

Prevents execution of code in data-only memory areas.

Different component of wireless network

Component

Explanation (Easy Language)

1. Wireless Access Point (WAP)

Connects wireless devices to a wired network. Works like a bridge.

2. Wireless Router

Routes data between devices and the internet.

3. Wireless Network Interface Card (NIC)

A hardware device (built-in or USB) that allows a computer or phone to connect to a wireless network.

4. Antennas

Used to send and receive radio signals. Can be internal or external.

Different component of wireless network

Component	Explanation
5. Clients (Devices)	Devices like smartphones, laptops, tablets, etc. that use the wireless network.
6. Modem	Connects to your ISP and brings internet into your network (often built into the router).
7. Network Protocols	Rules and standards (like Wi-Fi, Bluetooth, IEEE 802.11) that define how data is transmitted.
8. Authentication Server	Verifies users before allowing them to join the network (e.g., RADIUS server).

Traditional Techniques to Attack Wireless Networks

Attack Technique	Explanation
1. MAC Spoofing	Changing a device's MAC address to bypass filters or impersonate a trusted device.
2. Rogue Access Point	Setting up a fake Wi-Fi hotspot to trick users into connecting and stealing their data.
3. Evil Twin Attack	Creating a duplicate access point that looks like a legitimate one to intercept data.
4.. Denial of Service (DoS)	Flooding the network with fake traffic to disconnect or jam real users.
5. Eavesdropping	Listening to wireless communication to capture private data like passwords or messages.
6. WEP Cracking	Breaking weak encryption (like WEP) to gain access to the wireless network.

Identity theft:

Identity theft is a type of cybercrime in which someone illegally obtains and uses another person's personal information (like name, Aadhaar number, credit card details, or login credentials) without permission, usually to commit fraud or other crimes.

Types of Identity Theft:

1. Financial Identity Theft

- Stealing bank or credit card details to make unauthorized purchases or withdrawals.
- Example: Using someone's credit card to buy items online.

2. Criminal Identity Theft

- Using someone else's identity during an arrest or police investigation.
- Example: A criminal gives another person's name to avoid being caught.

3. Medical Identity Theft

- Using another person's identity to get medical treatment, prescription drugs, or insurance claims.
- Example: A fraudster uses your health insurance card for free treatment.

4. Synthetic Identity Theft

- Creating a fake identity using real and fake information (e.g., real Aadhaar number + fake name).

Child Identity Theft

- Using a child's identity (often unused credit history) to open accounts or get benefits.
- Often goes unnoticed for years until the child grows up.

Tax Identity Theft

- Filing a fake income tax return using someone else's PAN or identity to claim refunds.

Social Identity Theft

- Impersonating someone on social media to scam others, spread misinformation, or damage reputation.

Preventive Measures:

- Never share sensitive information online or over calls.
- Use strong, unique passwords.
- Enable two-factor authentication (2FA).
- Monitor financial and social media accounts regularly.

DIFFERENT TECHNIQUES OF IDENTITY THEFT

Technique	Description	How It Works	Example
1. Phishing	Tricking users into giving personal data through fake messages or emails	Fraudsters create fake websites or send emails that look like legitimate ones	Email claiming to be from a bank asking to “verify” your account
Skimming	Capturing card data using a hidden device	A skimmer is attached to ATMs or POS machines to read card information	Card data stolen at a petrol pump POS terminal
Data Breach	Hacking into databases of companies to steal stored user information	Attackers exploit system vulnerabilities to access and steal large user data	Hackers steal customer info from an e-commerce site
Shoulder Surfing	Observing someone entering sensitive info	Criminal watches over your shoulder while you type passwords or PINs	Someone watching you enter ATM PIN
Dumpster Diving	Retrieving personal documents from trash	Thieves go through garbage to find bills, bank statements, etc.	Stealing identity from thrown-away utility bills

DIFFERENT TECHNIQUES OF IDENTITY THEFT

Technique	Description	How It Works	Example
Social Engineering	Manipulating people to reveal private info	Attackers pretend to be trusted individuals (e.g., tech support, bank staff)	“Bank official” calls asking for OTP or card number
Public Wi-Fi Snooping	Intercepting data over insecure Wi-Fi networks	Hackers use sniffing tools to capture login details and other sensitive data	Using free airport Wi-Fi to steal social media credentials
Fake Websites/Apps	Creating fake versions of real websites or mobile apps	Users are tricked into logging in or entering payment details on fake sites	Fake SBI app stealing login details
Malware/Keyloggers	Using software to steal information or track keystrokes	Malware records what you type or sends info to hackers	Keylogger records email and banking passwords

Method	Description
Botnet Detection	Use of network monitoring tools to detect unusual traffic patterns and behaviors
Sinkholing	Redirects botnet communication to a controlled server to block attacker access
Takedown Operations	Collaborate with ISPs, domain registrars, and law enforcement to shut down botnet servers
Command-and-Control (C&C) Interception	Hijacking or blocking communication between infected devices and the botmaster
Patch Management	Applying security patches to stop further infections or spread of the botnet
Blacklisting Domains/IPs	Blocking known botnet domains and IP addresses at firewalls or DNS level
Public Awareness	Educating users about phishing, suspicious downloads, and safe online behavior

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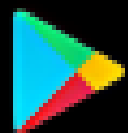


Gateway Classes

Empowering Learners, Transforming Futures



Thank You

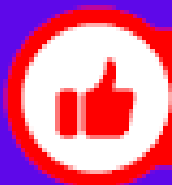


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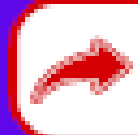
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